

Type 2 Diabetes: Glucose and HbA1c Monitoring

This is a synthetic, simplified reference document created for a portfolio software project. It is not real clinical guidance and must not be used for patient care.

Adults with type 2 diabetes should have fasting plasma glucose and hemoglobin A1c (HbA1c) checked at diagnosis and then at regular intervals thereafter. For patients meeting treatment goals with stable control, HbA1c testing every six months is generally sufficient. For patients with recent medication changes or inadequate control, testing every three months is recommended.

A fasting glucose reference range of 70-99 mg/dL is considered normal. Values persistently above 126 mg/dL on fasting testing are consistent with diabetes. An HbA1c target below 7.0% is a reasonable general goal for most non-pregnant adults, though targets should be individualized based on age, comorbidities, hypoglycemia risk, and disease duration.

Glucose values below 70 mg/dL indicate hypoglycemia and should prompt an immediate review of current medications, especially insulin and sulfonylureas. Values below 54 mg/dL represent clinically significant hypoglycemia requiring urgent attention. Severe hyperglycemia, generally above 400 mg/dL, or any glucose value accompanied by symptoms of diabetic ketoacidosis, warrants urgent clinical evaluation rather than routine follow-up.

Renal function should be assessed periodically in patients with diabetes, since several glucose-lowering medications, including metformin, require dose adjustment or temporary discontinuation in the setting of impaired kidney function or iodinated contrast exposure.